

Business Analytics Case Study

Why Customers Leave: A Study of Telco Customer Churn

AnalystLab Africa Data Analytics Internship Program

Week 5

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1. Introduction

Companies lose money when customers leave. Keeping a customer is cheaper than finding a new one. This report looks at data from a phone/internet company to find out why customers leave (this is called "churn") and which customers are most likely to leave. At the end, I give simple recommendations the company can use.

2. The Business Problem

The company wants to know:

- Why are customers leaving?
- Which customers are most likely to leave?

If the company knows this, it can act early and stop customers from leaving.

3. About the Dataset

The dataset has **7,043 customers** and **21 columns** of information, such as contract type, monthly charges and the services they use.

- **Churned customers:** 1,869 (26.5%)
- **Stayed:** 5,174 (73.5%)
- 11 rows had missing billing data and were removed before analysis.

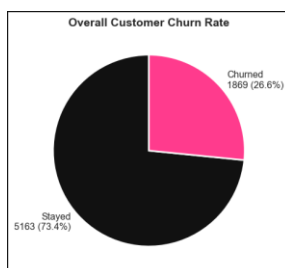


Figure 1: About 1 in 4 customers left the company.

4. What the Data Shows

Contract type matters most.

Customers on month-to-month contracts leave far more often (42.7%) than customers on one-year (11.3%) or two-year contracts (2.8%). Customers with no long contract can leave anytime, so they do.

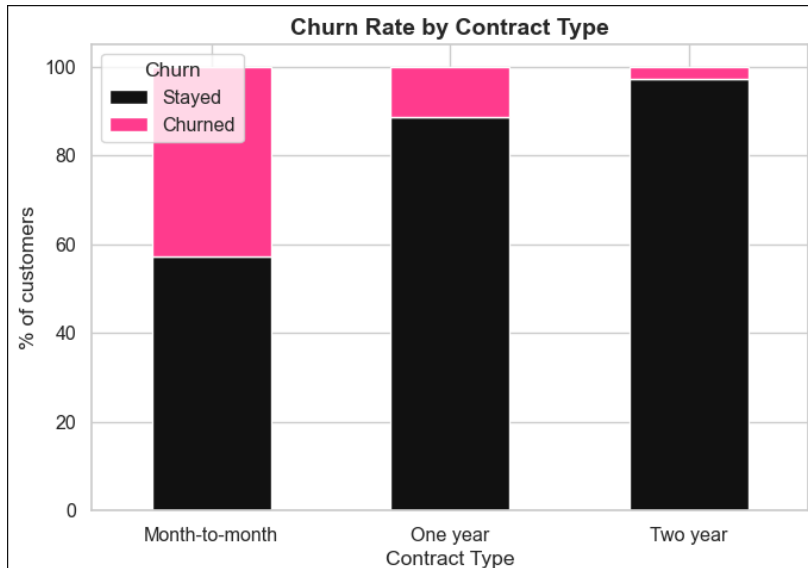


Figure 2: Month-to-month customers churn the most.

New customers leave more.

Customers who just joined are more likely to leave. Almost half (47.7%) of customers leave within their first year. After 4 years, only about 9.5% leave.

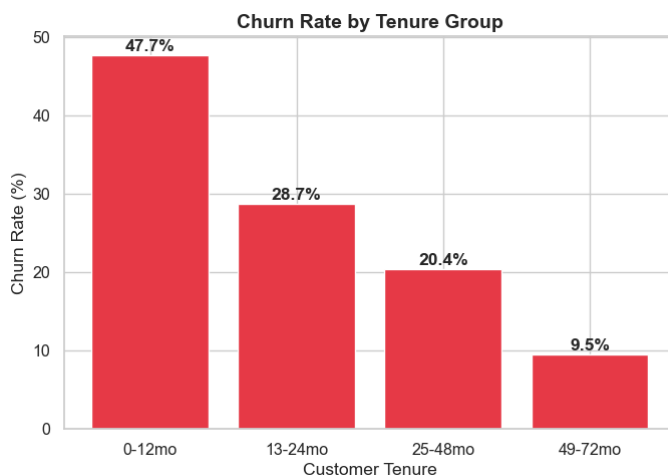


Figure 3: Churn drops as customers stay longer.

Fiber internet customers leave more.

Customers with fiber optic internet leave at 41.9%, more than double the DSL rate (19.0%) — even though fiber is the more expensive service. This suggests customers may not feel it's worth the price.

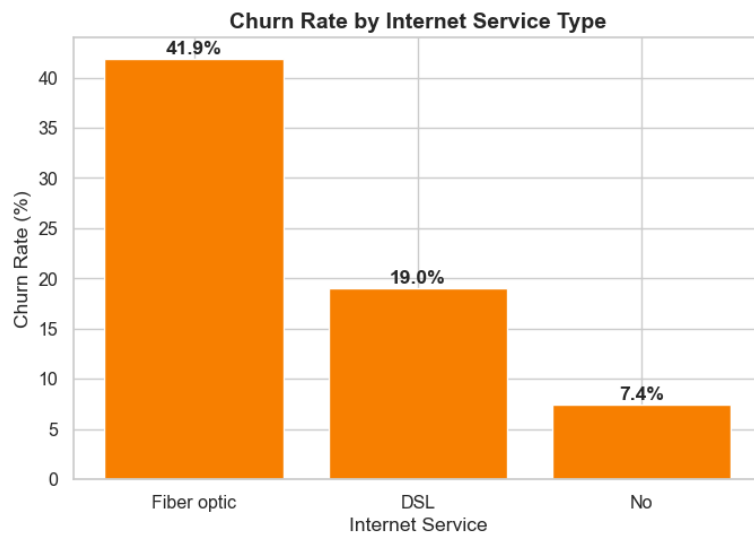


Figure 4: Fiber optic has the highest churn.

Monthly charges also matter.

Customers who leave tend to pay more per month than customers who stay.

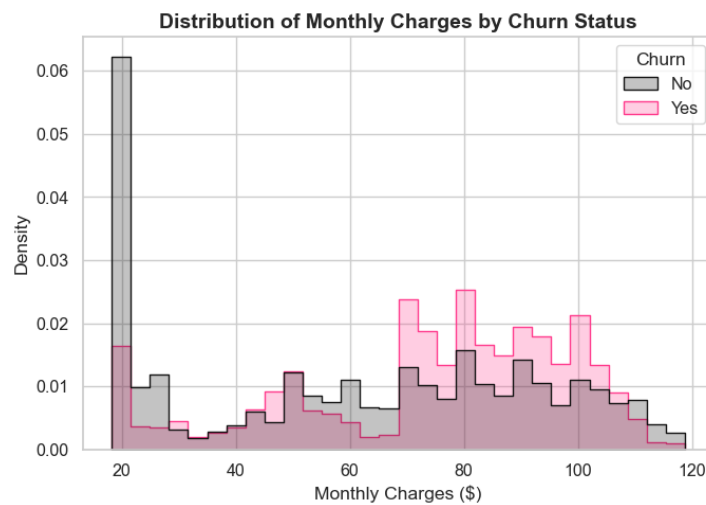


Figure 5: Customers who churn often pay higher monthly bills.

How you pay matters too.

Customers who pay by electronic check leave the most (45.3%). Customers who use automatic payments (bank transfer or card) leave much less (15–17%).

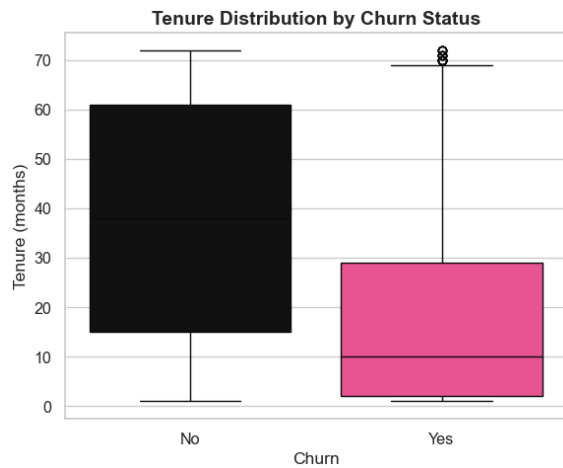


Figure 6: Electronic check users churn the most.

Support services help keep customers.

Customers without Tech Support or Online Security leave much more often (about 41–42%) than customers who have these services (about 15%).

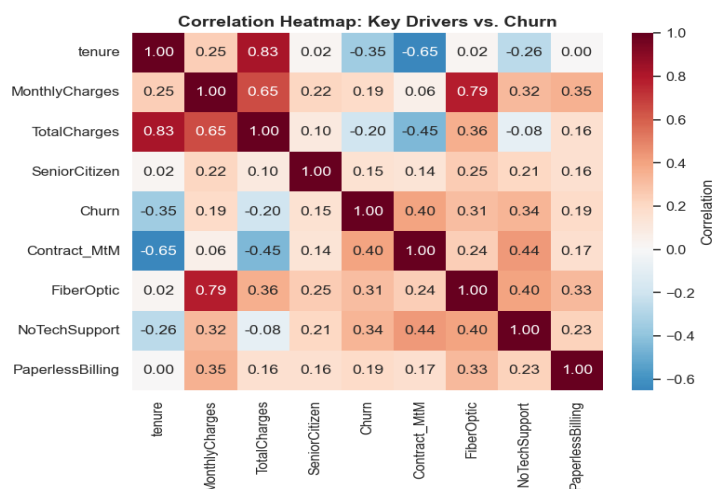


Figure 7: This chart confirms the same patterns using correlation numbers.

5. Key Insights

- Most churn happens in the **first year** of being a customer.
- **Month-to-month contracts** have the highest churn of any factor.
- **Fiber optic** customers pay more but leave more — something is not adding value for them.
- Customers who pay by **electronic check** are at high risk of leaving.
- Customers **without add-on services** (Tech Support, Online Security) are much more likely to leave.

- Customers who match **all three risk factors** — month-to-month, fiber optic, and electronic check — have a churn rate of **60.4%**, more than double the average. This is the highest-risk group.

6. Recommendations

1. **Offer discounts to move customers from month-to-month to yearly contracts.**
2. **Support new customers in their first year** with check-ins and small rewards at month 6 and month 12.
3. **Review the fiber optic service** — check pricing and service quality to see why customers are unhappy.
4. **Encourage customers to switch to automatic payments** instead of electronic check.
5. **Offer Tech Support and Online Security as a bundle** to new customers, since these reduce churn.
6. **Create a simple "risk score"** using these factors so the company can contact high-risk customers before they leave.

7. Conclusion

Customers don't leave randomly there is a clear pattern. Customers who are new, on month-to-month contracts, use fiber optic internet and pay by electronic check are the most likely to leave. This group alone represents a lot of lost revenue about \$139,000 lost per month or \$1.67 million per year, across all churned customers. By focusing on these specific groups, the company can reduce churn and protect its revenue.